

EMS Optimization Project — Final Status Report

Date: 2026-03-15
Test Status: 176/176 passing
Remote: Pushed to `origin/main`

Test Failure Analysis & Resolution

Root Cause

All 54 failures (45 FAILED + 9 ERROR) traced to a **single root cause**: a column name mismatch in `data/processed/demand_lambda_precinct.csv`.

Location	Expected Column	Actual Column
<code>src/ems_readiness/demand/arrival_generator.py:111</code>	<code>lambda_per_hour</code>	<code>crash_rate_per_hour</code>
<code>src/ems_readiness/optimization/allocator.py:116</code>	<code>lambda_per_hour</code>	<code>crash_rate_per_hour</code>

The data pipeline (`scripts/generate_all_data.py`) correctly generates the precinct lambda table with the column name `crash_rate_per_hour` , but two source modules still referenced the old column name `lambda_per_hour` . This caused a `KeyError` whenever:

- The NHPP arrival generator was constructed via `from_tables()` (affects all simulation & demand tests)
- The EMS allocator loaded demand data (affects optimization integration tests)

Affected Test Modules

Module	Failures	Errors	Type
test_demand_advanced.py	0	9	KeyError in fixture setup
test_simulation_core.py	14	0	KeyError during EMSSimulation.__init__
test_dispatch_logic.py	3	0	Same cascade
test_extreme_cases.py	10	0	Same cascade
test_integration.py	6	0	Same cascade
test_performance.py	3	0	Same cascade
test_regression.py	3	0	Same cascade
test_reproducibility.py	6	0	Same cascade

Fix Applied

Two one-line changes:

```
# arrival_generator.py line 111
- zip(precinct["precinct"].astype(int), precinct["lambda_per_hour"])
+ zip(precinct["precinct"].astype(int), precinct["crash_rate_per_hour"])

# allocator.py line 116
- demand = dl.set_index(dl["precinct"].astype(str))["lambda_per_hour"]
+ demand = dl.set_index(dl["precinct"].astype(str))["crash_rate_per_hour"]
```

Current Test Status

```
===== 176 passed in 2.86s =====
```

All 176 tests pass across 12 test modules:

Module	Tests	Status
test_demand_advanced.py	9	All pass
test_dispatch_logic.py	3	All pass
test_extreme_cases.py	10	All pass
test_integration.py	6	All pass
test_optimization_advanced.py	22	All pass
test_performance.py	3	All pass
test_properties.py	22	All pass
test_regression.py	3	All pass
test_reproducibility.py	6	All pass
test_seed_manager.py	9	All pass
test_service_advanced.py	39	All pass
test_simulation_advanced.py	16	All pass
test_simulation_core.py	28	All pass

Data Pipeline Status

make clean-data && make data completes successfully in ~65 seconds, generating all 11 processed files:

- firehouses_clean.csv , firehouses_manhattan.csv
 - precincts_manhattan.geojson
 - crashes_manhattan.csv , crashes_manhattan.parquet
 - demand_lambda_hourly.csv , demand_lambda_dow.csv , demand_lambda_precinct.csv
 - demand_model_summary.json
 - distance_matrix_firehouse_precinct.csv , distance_matrix_firehouse_precinct_manhattan.csv
-

Git Commit History (recent)

```
aed09cd fix: resolve column name mismatch (lambda_per_hour -> crash_rate_per_hour) --  
all 176 tests pass  
e79fefb feat: add 5 pipeline enhancements -- validation, progress bars, caching, par-  
allel processing, versioning  
aac03c9 feat: implement data dependency migration -- pipeline, auto-gen, clean Git  
6ace59d docs: add data dependency analysis and migration plan  
35ea7a5 feat: engineering hygiene - expanded tests, seed management, documentation  
f18fd53 Remove remaining emojis from scripts and notebooks
```

Remote Repository Status

- **Remote:** origin -> github.com/cnsp/ems-optimization.git
 - **Branch:** main
 - **Status:** Up to date (all commits pushed)
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Recommendations

1. **CI/CD:** Consider adding a GitHub Actions workflow to run `pytest tests/ -v` on every push/PR to catch regressions early.
2. **Column naming convention:** Standardize on either `lambda_per_hour` or `crash_rate_per_hour` project-wide. The current choice (`crash_rate_per_hour`) is more descriptive and domain-appropriate.
3. **Data contract tests:** Add a lightweight test that validates CSV column names match what the code expects, to prevent this class of bug from recurring.